Quadratics & Nonlinear Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Factoring Quadratic Equations

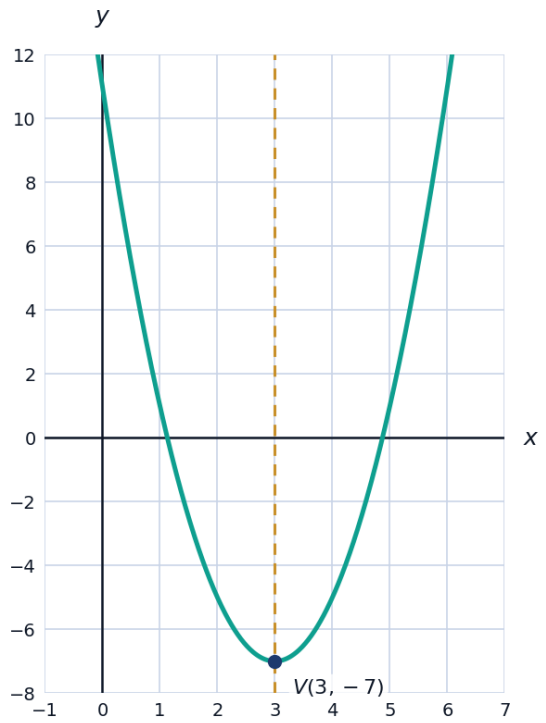
Factor each quadratic equation completely and find all real roots:

a $2x^2 - 11x + 12 = 0$

b $9x^2 - 25 = 0$

2 [Foundational] • [FRQ] • Target: Completing the Square & Vertex Form

Rewrite $f(x) = 2x^2 - 12x + 11$ in vertex form $f(x) = a(x - h)^2 + k$ by completing the square, and state the coordinates of its vertex and the equation of its axis of symmetry.



3 [Foundational] • [FRQ] • Target: Quadratic Formula & Simplifying Radicals

Use the quadratic formula to find the exact solutions of $3x^2 + 4x - 5 = 0$ in simplest radical form.

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4 [Foundational] • [FRQ] • Target: Discriminant & Nature of Roots

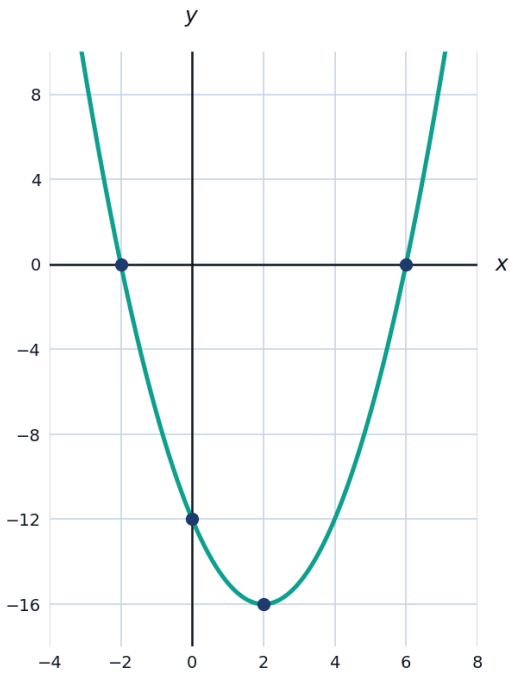
Compute the discriminant $\Delta = b^2 - 4ac$ for each quadratic equation and state the number of distinct real solutions:

a $4x^2 - 12x + 9 = 0$

b $2x^2 - 3x + 5 = 0$

5 [Foundational] • [FRQ] • Target: Factored Form & Symmetry of Parabolas

A parabola has x-intercepts at $(-2, 0)$ and $(6, 0)$ and passes through $(0, -12)$. Write the equation of the parabola in factored form and standard form, and find its minimum value.



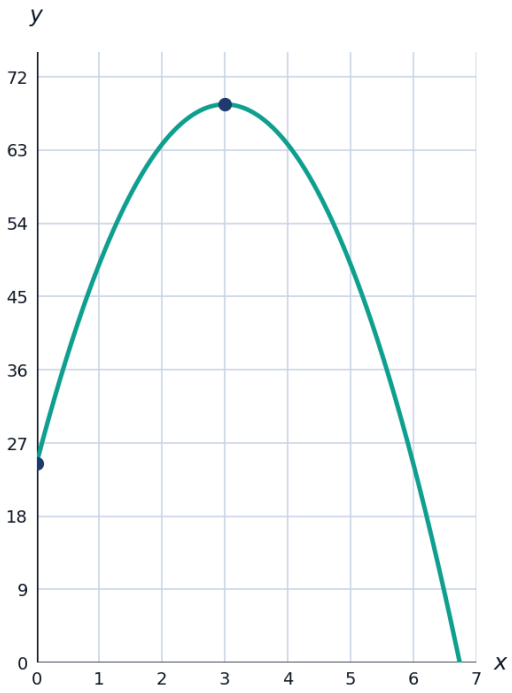
6 [Foundational] • [FRQ] • Target: Vieta's Formulas for Sum & Product of Roots

Without solving for the roots r_1 and r_2 of $5x^2 - 15x - 9 = 0$, use Vieta's formulas to find (a) $r_1 + r_2$, (b) $r_1 r_2$, and (c) $r_1^2 + r_2^2$.

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7 [Foundational] • [FRQ] • Target: Projectile Motion Quadratic Modeling

A flare is launched vertically upward from a platform $24\,\text{m}$ above the ground with initial velocity $32\,\text{m/s}$, so its height (in meters) after t seconds is $h(t) = -4.9t^2 + 29.4t + 24.5$. Find (a) the time t at which the flare reaches its maximum height, (b) the maximum height, and (c) the time when it hits the ground.



Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Discriminant Condition for Tangency

For what positive value of k does the quadratic equation $3x^2 - kx + 12 = 0$ have a single repeated real solution, and what is that solution?

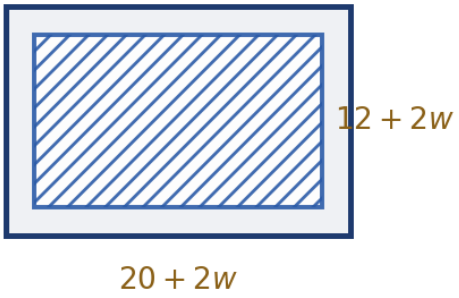
9 [Intermediate] • [FRQ] • Target: Equations Quadratic in Form

Solve the bi-quadratic equation $x^4 - 13x^2 + 36 = 0$ for all real solutions x.

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10 [Intermediate] • [FRQ] • Target: Geometric Area Quadratic Modeling

A rectangular garden of dimensions $12\,\text{m}$ by $20\,\text{m}$ is surrounded by a gravel walkway of uniform width w meters. If the total combined area of the garden and walkway is $384\,\text{m}^2$, find the width w of the walkway.

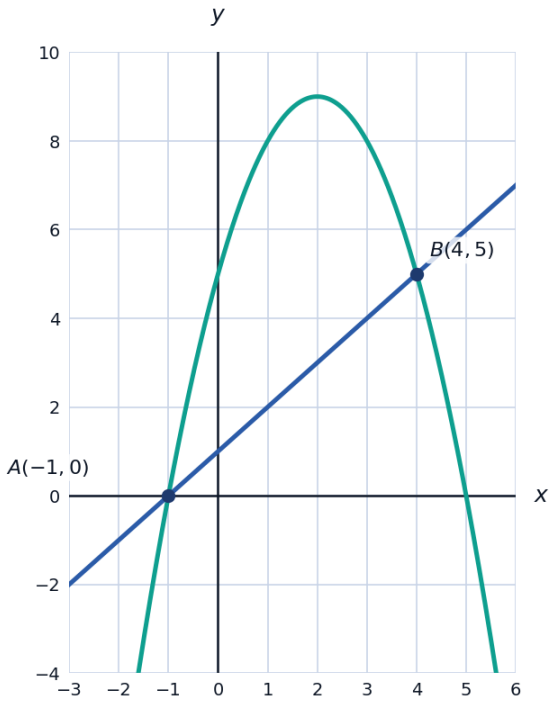


11 [Intermediate] • [FRQ] • Target: Revenue Optimization via Quadratic Vertex

A theater currently sells 300 tickets per night at 20 dollars per ticket. Market research shows that for every 1-dollar increase in ticket price, nightly attendance drops by 10 tickets. Express nightly revenue $R(x)$ as a quadratic function of the number of 1-dollar price increases x, and find the ticket price that maximizes revenue.

12 [Intermediate] • [FRQ] • Target: Parabola-Line Intersection

The parabola $y = -x^2 + 4x + 5$ and the line $y = x + 1$ intersect at two points A and B, as shown. Find the exact coordinates of A and B.



13 [Intermediate] • [FRQ] • Target: Root Difference & Vieta's Relations

If the roots of $x^2 - 6x + c = 0$ differ by 4, find the value of the constant c and both roots.

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14 [Intermediate] • [FRQ] • Target: Solving Quadratic Inequalities

Solve the quadratic inequality $2x^2 - 7x - 15 \leq 0$ and express the solution set in interval notation.



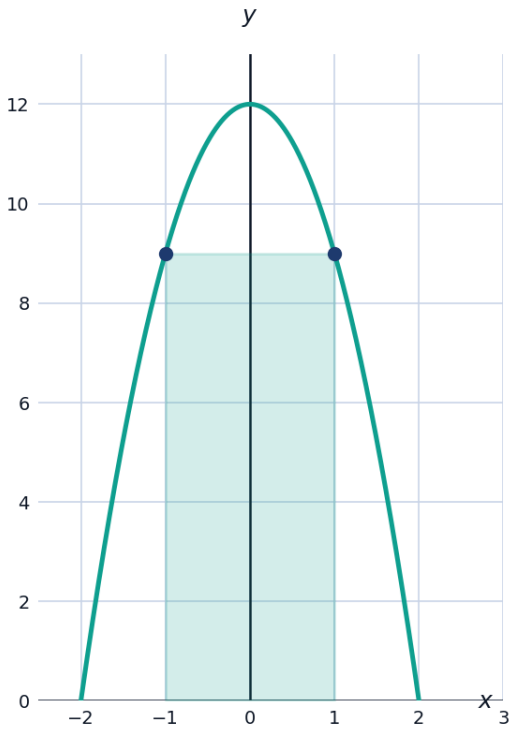
Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Symmetric Functions of Quadratic Roots • Cross-Lesson: Topic 10.01 (Rational Expressions & Vieta's Theorem)

Let r and s be the roots of $2x^2 - 8x + 5 = 0$. Without solving for r and s individually, find the monic quadratic equation whose roots are $\frac{1}{r}$ and $\frac{1}{s}$, and compute $\frac{r}{s} + \frac{s}{r}$.

16 [Challenging] • [FRQ] • Target: Inscribed Rectangle in a Parabolic Arch • Cross-Lesson: Topic 8.01 (Area & Perimeter of Inscribed Polygons)

A rectangle is inscribed under the arch of the parabola $y = 12 - 3x^2$ with its base on the x-axis. If the upper right vertex of the rectangle is at (x, y) with $x = 1$, find the area and perimeter of the rectangle; then determine whether $x = 1$ gives a larger area than $x = \frac{3}{2}$.



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17 [Challenging] • [FRQ] • Target: Parametric Line-Parabola Non-Intersection • Cross-Lesson: Topic 2.01 (Compound & Quadratic Inequalities)

For what range of values of the constant m does the line $y = mx - 5$ NOT intersect the parabola $y = x^2 - 2x + 4$?

18 [Challenging] • [FRQ] • Target: Pythagorean Theorem & Quadratic Algebraic Identity • Cross-Lesson: Topic 7.01 (Right Triangles & Pythagorean Theorem)

In right triangle ABC with $\angle C = 90^\circ$, the hypotenuse has length $13\,\text{cm}$ and the sum of the two legs is $17\,\text{cm}$. Set up a quadratic equation to find the lengths of both legs and the area of $\triangle ABC$.

19 [Challenging] • [FRQ] • Target: Exponential Equations Reducible to Quadratics • Cross-Lesson: Topic 4.01 (Exponential Functions & Laws of Exponents)

Solve the exponential-quadratic equation $4^x - 10 \cdot 2^x + 16 = 0$ for all real values of x.

20 [Challenging] • [FRQ] • Target: Structural Parabolic Suspension Modeling • Cross-Lesson: Topic 1.01 (Mathematical Modeling & Evaluation)

A suspension bridge cable hangs in the shape of a parabola $y = ax^2 + c$, where x is the horizontal distance (in meters) from the center of the span and y is the height (in meters) above the road deck. At the center $(x = 0)$, the cable is $6\,\text{m}$ above the deck; at the towers $x = \pm 60\,\text{m}$, the cable is $42\,\text{m}$ above the deck. Find a and c, and determine the height of a vertical suspender cable located $30\,\text{m}$ from the center.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Solving Quadratics by Factoring

What is the positive solution to the equation $2x^2 - 5x - 12 = 0$?

a. $\frac{3}{2}$

b. 3

c. 4

d. 6

22 [Foundational] • [MCQ] • Target: Identifying Vertex Coordinates

In the xy-plane, the parabola $y = -3(x + 4)^2 + 9$ has its vertex at (h, k). What is the value of $h + k$?

a. 5

b. 6

c. 13

d. -5

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23 [Foundational] • [MCQ] • Target: Equivalent Forms of Quadratic Functions

Which of the following functions is equivalent to $f(x) = x^2 + 8x + 7$?

a. $f(x) = (x + 4)^2 - 9$

b. $f(x) = (x + 4)^2 + 7$

c. $f(x) = (x + 8)^2 - 57$

d. $f(x) = (x + 4)^2 - 16$

24 [Foundational] • [MCQ] • Target: Sum of Solutions via Vieta's Formula

What is the sum of the solutions to the equation $3x^2 - 18x + 7 = 0$?

a. -6

b. $\frac{7}{3}$

c. 6

d. 18

25 [Foundational] • [MCQ] • Target: Using the Discriminant

How many distinct real solutions does the equation $5x^2 - 4x + 2 = 0$ have?

a. Zero

b. Exactly one

c. Exactly two

d. Infinitely many

26 [Foundational] • [MCQ] • Target: Solving by Square Roots

If $(x - 3)^2 = 20$, what is the greater of the two real solutions for x?

a. $3 + 2\sqrt{5}$

b. $3 + 4\sqrt{5}$

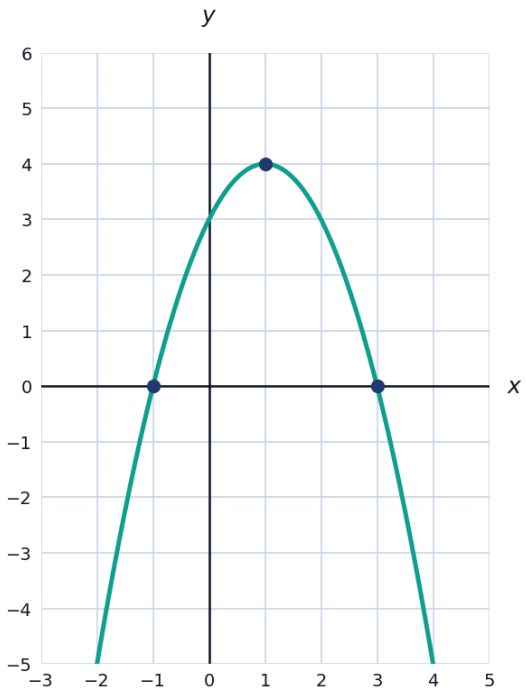
c. $-3 + 2\sqrt{5}$

d. 23

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27 [Foundational] • [MCQ] • Target: Graphical Interpretation of a and Δ

The graph of $y = ax^2 + bx + c$ is shown below. Which of the following correctly describes the signs of a and the discriminant $\Delta = b^2 - 4ac$?



a. $a > 0$ and $\Delta > 0$

b. $a > 0$ and $\Delta < 0$

c. $a < 0$ and $\Delta > 0$

d. $a < 0$ and $\Delta = 0$

28 [Foundational] • [MCQ] • Target: Reconstructing Quadratics from Roots

Which quadratic equation has solutions $x = -5$ and $x = \frac{2}{3}$?

a. $3x^2 + 13x - 10 = 0$

b. $3x^2 - 13x - 10 = 0$

c. $3x^2 + 17x + 10 = 0$

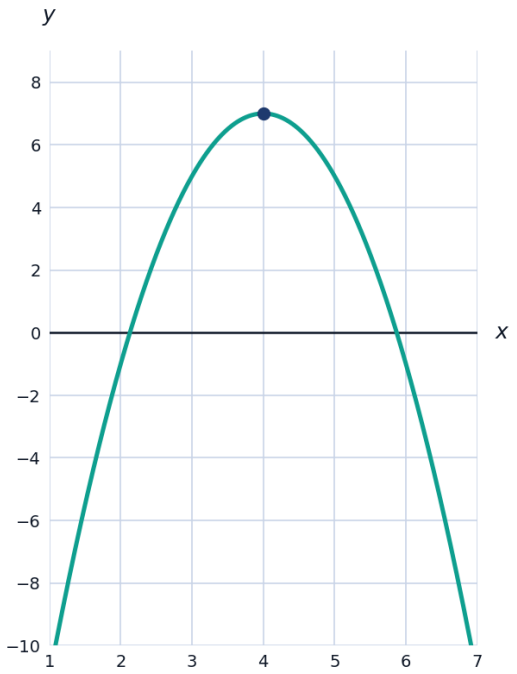
d. $3x^2 - 17x + 10 = 0$

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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29 [Intermediate] • [MCQ] • Target: Maximum Value of a Quadratic Function

The function $f(x) = -2x^2 + 16x - 25$ reaches its maximum value at $x = a$. What is the maximum value $f(a)$?



a. 4

b. 7

c. 9

d. 11

30 [Intermediate] • [MCQ] • Target: Discriminant Inequalities

In the quadratic equation $kx^2 - 12x + 9 = 0$, k is a nonzero constant. If the equation has no real solutions, which inequality must k satisfy?

a. $k < 4$

b. $k > 4$

c. $k < -4$

d. $0 < k < 4$

31 [Intermediate] • [MCQ] • Target: Relating Vertex Form to Standard Coefficients

If the parabola $y = x^2 + bx + c$ has vertex $(3, -4)$, what is the value of $b + c$?

a. -1

b. 1

c. 5

d. 11

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32 [Intermediate] • [MCQ] • Target: Axis of Symmetry from Equal Outputs

A quadratic function $g(x)$ satisfies $g(-1) = 8$ and $g(7) = 8$. What is the x-coordinate of the vertex of the graph of $y = g(x)$?

a. 2

b. 3

c. 4

d. 6

33 [Intermediate] • [MCQ] • Target: Product of Real Roots in Reducible Quartics

What is the product of all real solutions to $(x^2 - 5x)^2 - 2(x^2 - 5x) - 24 = 0$?

a. -24

b. -12

c. 12

d. 24

34 [Intermediate] • [MCQ] • Target: Structure in Quadratic Forms (Vertex vs. Factored)

A ball is thrown upward so that its height h, in feet, after t seconds is $h(t) = -16t^2 + 64t + 80$. Which equivalent form of $h(t)$ displays the time at which the ball hits the ground as a constant or coefficient?

a. $h(t) = -16(t - 2)^2 + 144$

b. $h(t) = -16(t - 5)(t + 1)$

c. $h(t) = -16t(t - 4) + 80$

d. $h(t) = -8(2t^2 - 8t - 10)$

35 [Intermediate] • [MCQ] • Target: Midpoint of a Secant Chord on a Parabola

If the graphs of $y = x^2 - 5x + 7$ and $y = 2x - 3$ intersect at (x_1, y_1) and (x_2, y_2), what is the midpoint of the segment connecting the two intersection points?

a. $(3.5, 4)$

b. $(3.5, 5)$

c. $(2.5, 2)$

d. $(5, 7)$

36 [Intermediate] • [MCQ] • Target: Fitting a Quadratic Through Three Points

For all real x, the function $f(x) = ax^2 + bx + c$ satisfies $f(0) = 3$, $f(1) = 2$, and $f(2) = 5$. What is the value of $f(4)$?

a. 19

b. 21

c. 23

d. 27

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37 [Intermediate] • [MCQ] • Target: Interval Length of Quadratic Inequalities

What is the length of the interval of all real x satisfying $x^2 - 4x - 21 \leq 0$?

a. 4

b. 7

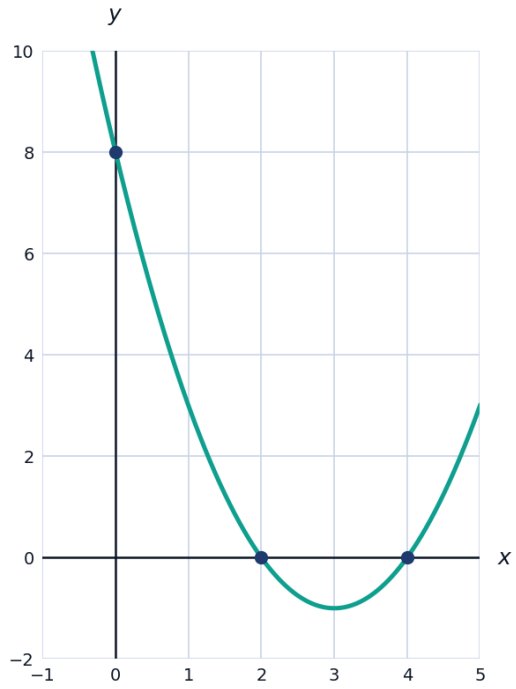
c. 10

d. 14

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Triangle Area from Parabolic Intercepts • Cross-Lesson: Topic 7.01 (Coordinate Triangle Area)

In the xy-plane, the parabola $y = x^2 - 6x + 8$ intersects the x-axis at A and B and the y-axis at C. What is the area of $\triangle ABC$?



a. 6

b. 8

c. 12

d. 16

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39 [Challenging] • [MCQ] • Target: Cubic Power Sums of Quadratic Roots • Cross-Lesson: Topic 4.01 (Polynomial Identities)

If r and s are the roots of $x^2 - 5x + 3 = 0$, what is the value of $r^3 + s^3$?

a. 80

b. 95

c. 110

d. 125

40 [Challenging] • [MCQ] • Target: Radical Equations & Extraneous Roots • Cross-Lesson: Topic 4.01 (Radicals & Rational Exponents)

For how many real values of x is $\sqrt{2x + 15} = x$?

a. Zero

b. One

c. Two

d. Three

41 [Challenging] • [MCQ] • Target: Circle-Parabola Intersection Trapezoid • Cross-Lesson: Topic 9.01 (Circles) & Topic 8.01 (Trapezoid Area)

A circle with equation $x^2 + y^2 = 13$ and a parabola with equation $y = x^2 - 7$ intersect at four points in the xy-plane. What is the area of the trapezoid formed by these four intersection points?

a. 15

b. 20

c. 25

d. 30

42 [Challenging] • [MCQ] • Target: Reconstructing Quadratic Inequalities from Intervals • Cross-Lesson: Topic 2.01 (Inequality Solution Intervals)

If the quadratic inequality $ax^2 + bx + 6 > 0$ has solution set $(-1, 3)$, what is the value of $a + b$?

a. -2

b. 0

c. 2

d. 4

43 [Challenging] • [MCQ] • Target: Constrained Optimization on a Closed Interval • Cross-Lesson: Topic 2.01 (Domain Restrictions)

Let $f(x) = x^2 - 4x + 7$. What is the minimum value of $f(x)$ over the restricted interval $3 \le x \le 6$?

a. 3

b. 4

c. 7

d. 19

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44 [Challenging] • [MCQ] • Target: Arithmetic & Harmonic Means via Quadratics • Cross-Lesson: Topic 6.01 (Statistics & Means)

Two positive numbers have a harmonic mean of 4.8 and an arithmetic mean of 5. What is the positive difference between the two numbers?

a. 1

b. 2

c. 3

d. 4

45 [Challenging] • [MCQ] • Target: Vertex Distance in the Coordinate Plane • Cross-Lesson: Topic 7.01 (Distance Formula & Pythagorean Triples)

In the xy-plane, what is the shortest distance from the vertex of the parabola $y = 2(x - 3)^2 + 4$ to the origin $(0, 0)$?

a. 5

b. $\sqrt{13}$

c. 7

d. 25

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